

Defense notes — Imagining the Target

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**20 minutes

1. Title

0:20 · cum 0:20

Good morning everyone, and thank you for being here.

2. Composed Image Retrieval

1:05 · cum 1:25

Imagine that you see a chair online that you really like, but you would like to find a dining set that includes that same chair design together with a table.

Searching only with the image of the chair would mainly return the same chair or other visually similar chairs. It would not understand that you are looking for it as part of a dining set.

On the other hand, searching only for "a dining table and chairs" would return many different sets, but probably not the specific chair you selected.

To express exactly what you are looking for, you need to combine both elements: the image identifies the chair, while the text describes the context in which you want to find it.

This is an example of composed image retrieval, which is the task I worked on in my thesis. The query consists of a reference image and a modification text, and the goal is to retrieve images that preserve the reference object while also satisfying the modification described by the text.

*One naming note: the reference image and modification text are simply the two parts of the query, so from here on I will call them the **query image** and the **query text**.*

3. Instance-level composed retrieval

0:55 · cum 2:20

My thesis focuses specifically on the instance-level version of this task.

In category-level composed image retrieval, any object belonging to the correct class may satisfy the query. For example, if the query asks for a chair around a dining table, any suitable chair design might be considered correct.

In instance-level retrieval, this is not enough. The system must retrieve the exact same chair shown in the query image, while also satisfying the query text.

So the main challenge is to detect that one specific instance among many look-alikes of the same class — even when its context, appearance, viewpoint, or visual domain changes.

4. The i-CIR benchmark

1:05 · cum 3:25

For our experiments we use the i-CIR benchmark, which was specifically designed for instance-level composed image retrieval.

Each object instance is associated with several composed queries, and every composed query consists of a query image and a query text.

Importantly, each query may have multiple correct target images — several database images can correctly depict the same object under the requested modification. So evaluation is not about finding one right answer; it is about ranking all of them highly.

In addition, each instance has its own database of carefully curated hard negatives. These are designed to confuse the system. Some are visually similar to the query image but fail to satisfy the query text. Others satisfy the query text but depict a different object instance. Some match both parts partially while still being incorrect. This prevents the system from succeeding by relying only on the image or only on the text.

5. The i-CIR benchmark — scale

0:35 · cum 4:00

A word on scale. i-CIR contains a database of seven hundred fifty-two thousand images, two hundred and two object instances, and one thousand eight hundred eighty-three composed queries, with a median of six queries per instance.

The histograms show the distributions: image and text queries per instance, hard negatives per instance, and the number of correct targets per composed query — which confirms that many queries have more than one ground truth.

6. Problem formulation — a zero-shot setting

0:55 · cum 4:55

Formally, we are given a query image and a query text, and we want to retrieve database images that are relevant to both.

In this thesis we adopt a zero-shot setting: we do not train or fine-tune any model on the i-CIR dataset.

Why does this matter? Supervised composed image retrieval requires labelled triplets — a query image, a query text, and a correct target image. At the instance level, collecting these annotations for every new object would be expensive and hard to scale. Worse, a model trained on a fixed set of instances does not necessarily generalize to instances it has never seen — and in a real system new objects appear constantly. So a supervised approach would decrease both the scalability and the generalizability of the system.

Instead, by using frozen pre-trained models, the retrieval system applies directly to previously unseen object instances, with no additional training.

7. Retrieval, ranking, and the metric

0:55 · cum 5:50

First, how retrieval actually works. Given a query, the system assigns a relevance score to every database image, and sorts them from most to least relevant. This ranked list is the output; the evaluation checks where the correct targets land in it.

For a single query we use Average Precision. AP is high when the correct images appear near the top of the ranking; if they appear lower, the score decreases.

We then compute mean Average Precision by averaging AP across all queries.

However, the number of queries is not the same for every instance. If we averaged directly over all queries, instances with more queries would dominate the result.

For this reason our main metric is macro-mAP: we first average within each object instance, and then across instances — so every instance contributes equally.

8. Vision-language models: CLIP and SigLIP

1:10 · cum 7:00

The main technical challenge is combining the information from the query image and the query text. To represent both modalities we use frozen, pre-trained vision-language models: CLIP and SigLIP.

These use a dual-encoder, or two-tower, architecture. One encoder processes images, the other processes text, and both map their inputs into the same embedding space. Because the representations share this space, we measure semantic similarity with cosine similarity: an

image scores high when its embedding is close to the embedding of the corresponding text.

The models are used completely frozen — we only extract embeddings, we never update their parameters.

CLIP and SigLIP follow the same dual-encoder principle but differ in training. CLIP compares all image-text pairs within a batch, using a softmax contrastive objective. SigLIP instead classifies each pair independently with a sigmoid loss — no global softmax, which scales better to large batches.

We compare these frozen backbones to test whether our method depends on one particular vision-language model.

9. The baseline: BASIC

1:10 · cum 8:10

The baseline for our work is BASIC, the zero-shot method introduced together with the i-CIR benchmark by Psomas and colleagues in 2025.

BASIC processes the two parts of the composed query separately. The query image is encoded by the visual encoder, and the query text by the text encoder. The database images are encoded in advance by the visual encoder.

For every database image, BASIC computes two similarity scores: the visual similarity, how similar the database image is to the query image; and the textual similarity, how well it matches the query text. Both are dot products between the corresponding embeddings.

The two scores are then multiplied. This multiplicative fusion acts like a soft logical AND: a database image receives a high final score only if it matches both the query image and the query text.

The method also includes additional processing steps — centering, score normalization, contextualization — which I will discuss later.

10. Limitation of the baseline — our proposal

1:10 · cum 9:20

The main limitation of the baseline is that the query image and the query text are never processed jointly. The image branch only sees the query image, the text branch only sees the query text. As a result the model does not explicitly represent the complete composed query.

To address this, we introduce a third scoring branch based on a generated target caption.

This is the full pipeline. The query image and query text go, as before, to the image and text branches. In addition, together, they go to a multimodal language model that imagines the target and writes a caption of it. That caption is embedded by the same frozen text encoder and becomes a third similarity score.

So for every database image we now compute three scores — to the image, to the text, and to the generated caption — and combine them. I will detail the captioner and the fusion on the next slides.

11. The captioner: InternVL 3.5-8B

0:50 · cum 10:10

For target-caption generation we use the frozen eight-billion-parameter version of InternVL 3.5 — an open-source multimodal LLM that can jointly process an image and text, which is exactly what the caption branch requires. At eight billion parameters it is a practical balance of quality and cost, and we use it entirely without fine-tuning.

Concretely, it receives the query image, the query text, and our instruction to describe the imagined target. In the chair query it sees the specific chair together with the text "placed around a table", and generates: "White metallic chair with scrolled armrests placed around a

table."

Notice that this single caption contains information from both parts of the query — the visual identity of the chair and the context specified by the text. That is the signal the two separate branches cannot represent on their own. We embed it with the same frozen text encoder to get the third similarity score.

12. Designing the instruction

0:45 · cum 10:55

An important part of the caption-generation stage is the instruction given to the multimodal model.

The quality of the generated caption depends strongly on how the task is formulated. For this reason I carried out an extensive prompt-design process and tested more than twenty different instructions, varying the level of detail, the use of examples, the order of the constraints, and restrictions intended to prevent the model from introducing irrelevant information.

Based on these experiments I selected five complementary prompting strategies. Each describes the imagined target from a slightly different perspective, and together they produce a diverse set of captions whose embeddings we average.

| *What each strategy does → backup 31.*

13. Effect of the caption branch

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Adding a single generated caption improves macro-mAP by about 5.9 points, from 17.95 to 23.83. This suggests the caption captures useful compositional information that is not fully represented by the image and text branches separately.

Using multiple captions improves the result further. We generate five captions with different prompting strategies, average their embeddings, and use the result for retrieval. This reduces the influence of prompt-specific wording and emphasizes information that is consistent across the captions.

With this five-caption average, macro-mAP reaches 25.41 — about 7.5 points above the original image-text baseline.

| *From this point on, all results use CLIP ViT-L/14 as the backbone, unless stated otherwise.*

14. Combining the three scores

1:00 · cum 12:50

We now have three similarity scores per database image — to the query image, the query text, and the generated caption. How do we combine them?

First, min-based normalization puts the three branches on comparable scales, and any negative residual values are clamped to zero — meaning they provide no positive evidence for the candidate.

The normalized scores are then combined multiplicatively, because a good result must match all three branches at the same time — like a soft logical AND.

On top of that we use a Harris-inspired penalty. The idea comes from the Harris corner detector, where a strong response in only one direction is not enough. Similarly, a candidate should not win just because it matches one branch very well while matching the others poorly — for example an image that shows the exact chair but not around a table, or chairs around a table that are not the specific chair. The Harris term penalizes this imbalance, weighted by a hyperparameter λ .

The surface on the right shows the effect: the score is only high when both arguments are simultaneously large.

15. Two refinements: centering and contextualisation

1:05 · cum 13:55

Two further techniques improve retrieval — one on the embeddings, one on the query text itself.

****Centering****, applied to the embeddings. Vision-language embedding spaces contain a large common component shared by many embeddings — generic visual or linguistic patterns rather than the distinctive information needed for retrieval. We compute a mean embedding for each modality on a separate corpus, subtract it, and re-normalise. This isolates the semantic content from the shared distributional bias.

****Contextualisation****, applied to the raw query text — so it is a query-side step, not a score post-processing step. The text encoder is trained mostly on full captions, so a bare fragment such as "sculpture" or "during sunset" is out-of-distribution and produces a text feature poorly aligned with image features. We therefore enrich the query with terms from a subject corpus, before and after — "dog during the sunset", "sculpture dog" — then embed and average the variants, giving a more robust textual feature.

16. Adding the steps on top of each other

0:40 · cum 14:35

Here we add each step cumulatively, for the baseline in blue and our caption pipeline in red, on CLIP ViT-L/14.

The caption pipeline stays above the baseline along the entire ladder. Centering gives a large jump for the baseline, and contextualisation gives the largest single gain.

If asked why the ladder stops here: two further BASIC steps — the contrastive projection and query expansion — do not help once the caption branch is present. Numbers → backup 24.

17. Does it transfer? Four frozen backbones

0:55 · cum 15:30

We evaluate the pipeline with four frozen backbones, spanning two model families and two capacity levels. CLIP ViT-L/14 is the backbone most commonly reported in the zero-shot CIR literature, which makes our numbers comparable. CLIP ViT-H/14 is the higher-capacity variant. SigLIP replaces the softmax contrastive loss with the pairwise sigmoid loss — we use SigLIP ViT-L/16 and the more recent SigLIP2 g/16, the best-aligned encoder of the four.

The caption branch improves every single one of them. The largest jump is on SigLIP-Large, where macro-mAP roughly doubles, from twenty-one to forty-three. SigLIP2 is the strongest base model.

This shows that our contribution does not depend on one particular vision-language model.

18. Our best overall system

0:50 · cum 16:20

To state the final system plainly: the SigLIP2 backbone, the caption branch with five averaged captions, and post-processing through centering and contextualisation. Every component is frozen.

It reaches 61.98 macro-mAP and 62.07 mAP. Beyond mAP: the correct target is ranked first for 71% of queries, and appears in the top ten for 95% of queries. Recall at one hundred is 94%.

That high recall is important — I will come back to it in the future work.

The SigLIP2 ladder that produces 61.98 → backup 25.

19. Qualitative example — success

0:30 · cum 16:50 (flex)

A qualitative success. The query is Homer Simpson, with the modification "as a plushie".

The plain image-times-text product scores only 4% AP — it drifts to generic plushies: SpongeBob, minions, a teddy bear. BASIC reaches 33%. Adding the generated caption — "Homer Simpson as a plushie, with the same facial features" — pins the retrieval to actual Homer plushies and lifts AP to 88%. This is the caption branch doing exactly what it is meant to do: keeping the specific identity while honouring the modification.

20. Qualitative example — failure

0:30 · cum 17:20 (flex)

And an honest failure. Gold metal eyeglasses, "worn by a dog".

Here all three methods fail — AP near zero. The single ground truth is a photo of a dog actually wearing these glasses. Every method instead returns glasses photographed on tables or held up, because the visual branch anchors hard on the glasses themselves and the caption cannot overcome a target this rare.

This is the characteristic failure mode: when the modification demands a drastic, rarely-photographed context, the image branch dominates and the correct target is simply not near the top.

21. Limitations and future work

0:50 · cum 18:10

Three limitations. The main cost is caption generation — about fifteen seconds per query, which dominates online inference. The results also depend on prompt design: averaging five captions mitigates this, but does not remove it. And the fixed, unweighted product cannot adapt when one branch should matter more than the others.

Several directions follow naturally. A stronger or more efficient captioner — or a single high-quality caption that matches the five-caption average without five generations — would cut the main compute cost. The fixed product could be replaced by a learned or weighted fusion of the three similarities, relaxing the zero-shot constraint in exchange for accuracy.

Because recall is already high — the correct instance lands in the top-ranked shortlist for the large majority of queries — a second-stage reranker that reorders this shortlist is a promising direction: most of the remaining error is in the ordering of candidates that are already retrieved. A text-conditioned reranker that reorders by similarity to the target caption, rather than to the query image, is a natural candidate, since it keeps the modification in the loop at the reranking stage.

Finally, the caption-as-third-modality idea is benchmark-agnostic and could be evaluated beyond i-CIR — on category-level CIR benchmarks and other instance-level retrieval settings.

22. Conclusions

0:40 · cum 18:50

Two conclusions.

First, the caption is the single most effective component. Adding the target caption improves every backbone on the full dataset.

Second, SigLIP2 is the strongest backbone for the pipeline, reaching the thesis-best 61.98 macro-mAP, fully zero-shot.

If pushed further: the caption alone already beats the plain image-times-text product on CLIP-L (backup 26), and centering and the caption are complementary — their gains stack (backup 27).

23. Thank you

Thank you. I am happy to take questions.

Q&A map

Full CLIP-L ablation, every rung (incl. projection and query expansion) → **backup 24**. SigLIP2 ladder (produces 61.98) → **backup 25**. Component contributions — image-only, text-only, caption-only, product, triplet — showing the caption alone beats the product → **backup 26**. Centering and the caption are complementary (gains stack) → **backup 27**. "The paper reports 34.35 for BASIC" → **backup 28** (our reproduction gives 32.48; the gap is in the image branch, from re-extracting features in a newer software stack; both methods are evaluated in the same environment, so the comparisons hold). Latency breakdown → **backup 29**. The five instruction strategies with their sample captions → **backup 30**; what each strategy does → **backup 31**.

Metric on slide 7; the full pipeline figure on slide 10; fusion detail on slide 14; the backbone table on slide 17.